

2024

COMPUTER SCIENCE

Paper : CSMC-102

(Data Structures and Algorithms)

Full Marks : 70

*The figures in the margin indicate full marks.**Candidates are required to give their answers in their own words as far as practicable.*Answer question nos. 1 and 2, and *any four* questions from the rest.

2×5

1. Answer *any five* questions :

- (a) Critically comment on the following : "Little-oh is used to describe a loose upper bound of an algorithm."
- (b) Justify whether " $O(\log_2 n) = O(\log_5 n)$ ". Hence, what we may conclude?
- (c) Why does the evaluation of an infix expression take longer time than the evaluation of its equivalent prefix or postfix expression?
- (d) State the vertex cover problem in its decision version and optimization version.
- (e) Compute the value of 54×95 using Karatsuba algorithm.
- (f) Relatively compare amortized analysis and average case analysis.
- (g) Encode the word 'Calcutta' using Huffman code.
- (h) Why is 'memorization' important when solving a problem using dynamic programming? Give an example.

2. Answer *any five* questions :

- (a) Define the term 'heuristic'. When a heuristic algorithm is usually devised? How is it different from meta-heuristic? 1+1+2.
- (b) Given an undirected, unweighted dense graph, a start vertex, and an end vertex. Your task is to extract the shortest path, if one exists, from the start vertex to the end vertex. Devise an efficient algorithm to solve this problem and judge its efficiency based on the appropriate data structure you employ. 4
- (c) Define binomial heap and give an example binomial heap that comprises 13 vertices. Trace an algorithm that constructs a binomial heap with the help of the following keys : 5, 9, 10, 1, 4, 7, 11, 3, 8, 2, 6. 2+2
- (d) Why is *QuickSort* so-called even after its *bad* worst-case computational complexity? How can be *randomized QuickSort* may work better? Mention two situations where *randomized QuickSort* may work worst. 1½+1½+1

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- (c) Relatively compare simulated annealing and simulated evolution as nature-inspired algorithms to solve hard problems. 4

- (f) Define the satisfiability problem (SAT). Consider the following instance of SAT and convert it into a corresponding instance of 3 SAT, thus proving that 3 SAT is also NP-complete. 1+3

$$\psi = (b' + x' + t + m' + r + q)(p + a + w')(h' + y + n' + s')(f + c')$$

- (g) State the Longest Common Subsequence (LCS) problem that we like to compute for a given pair of strings. Given two strings $S_1 = \text{"ABAABA"}$ and $S_2 = \text{"BABBBAB"}$, devise an algorithm based on tabulation to compute the LCS between S_1 and S_2 . What are the complexities of your algorithm? 1+2+1

- (h) Obtain the addressing formula for the element $A[i_1, i_2, \dots, i_n]$ in an array declared as $A[l_1 \dots u_1, l_2 \dots u_2, \dots, l_n \dots u_n]$. Assume a column-major array representation with one word per element and α is the address of $A[l_1, l_2, \dots, l_n]$. Show that the addressing formula is linear. 3+1

3. (a) What is the main idea behind designing a matrix multiplication algorithm using the divide and conquer technique? Perform $A \times B$ using Strassen's algorithm, where A and B are two distinct identical matrices of your choice.

- (b) Define a *matroid* and state its structural properties.

- (c) Consider the following graph, $G = (V, E)$, where $V = \{a, b, c, d, e, f, g\}$ and $E = \{(a, b), (a, c), (a, g), (b, e), (c, f), (e, d), (e, g), (f, g), (g, b), (g, d)\}$. Use a deterministic DFS to differentiate the edges of G , assuming 'a' as the start vertex. (1+4)+2+3

4. (a) What do you mean by AVL tree? Why is it necessary to construct?

- (b) Consider the following sequence of elements and insert them in the order given to build an AVL tree after each insertion. Then, delete 49 from the final AVL tree obtained. Clearly show all necessary rotations.

9, 79, 89, 19, 99, 49, 29, 59, 39, 69

- (c) What do you mean by tree traversal? Justify whether a binary tree could be uniquely generated if preorder and postorder traversals are given. Give an example to support your answer. (1+1)+(3+1)+(1+2+1)

5. (a) Define B-tree. Why is it necessary?

- (b) Insert the keys $b_1, h_1, f_1, a_1, j_1, d_1, h_2, n_2, i_2, e_1, s_1, i_1, r_1, x_1, c_2, n_1, n_3, t_1, u_1, p_1$ into an initially empty tree, in the order given, so that the computed B-tree is of order 5. Then, delete the keys one after another from the final B-tree obtained, in the following order: h_2, r_1, p_1, d_1 , and obtain subsequent B-trees. (1½+1)+(5+2½)

6. (a) Specify the graph and state when a graph is called strongly connected. What do you mean by *strongly connected component* (SCC)? Let (i) $G = (V, E)$ be a digraph, (ii) $G' = (V, E')$ be an SCC of G , and (iii) $v \in V'$ be its root. Then prove that $v \rightarrow^* w$ following tree-edges for all $w \in V'$ and v is the node with the largest DFS completion number in V' .

- ✓ (b) Give the mathematical definition of sorting. Briefly state the underlying principle of *GraphSort*, and utilize this principle to justify if it is a stable sorting algorithm with the help of the following sequence of numbers.

$$(1+1+3)+(1+4)$$

$$4, 3, 1, 4, 2, 5, 4, 1, 3$$

7. (a) What do you mean by approximation algorithm? Why is it necessary to devise? Show that a polynomial time computable approximation algorithm exists for the vertex cover problem, the performance ratio of which is as good as 2. Compute the computational complexity of this algorithm with justification.
- (b) Draw the Venn diagram comprising domains of problems P, NP, NP-complete, and NP-hard, and briefly explain.
- $$(2^{1/2}+1+2^{1/2}+1)+3$$
8. (a) Define *mergeable heaps* with respect to a set of associated operations and give examples of such heaps. Also, state what *Decrease-Key*(*H*, *x*, *K*) operation does.
- (b) Consider the operation of extracting the node with a minimum key belonging to the largest binomial tree of a binomial heap that comprises at least 20 nodes and at least four binomial trees, and clearly show the steps of this operation with necessary diagrams to obtain the resulting binomial heap after extraction.
- $$[(3+1)+1]+5$$
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